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Structural Re-Parameterization, Spatial Coordinate Encoding, and Cross-Domain Robustness for Real-Time Safety Helmet Detection in Construction Surveillance

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Summary

The paper proposes Rep-YOLO11s, an edge-focused detector for safety-helmet monitoring that combines three known ideas in a deployment-aware way: (i) structural re-parameterization in the neck to reduce memory access costs at inference, (ii) a CoordConv stem to inject elevation priors that help suppress floor-level distractors, and (iii) bi-level routing attention (BiFormer) in high-level neck stages to focus computation on small helmets. A decoupled training scheme uses Focal-EIoU with Task-Aligned Assignment to mitigate class imbalance. Empirically, the method attains real-time throughput on commodity GPUs, shows competitive

accuracy on SHWD, and reports stronger zero-shot transfer on the Hard Hat Workers (HHW) dataset, with extensive latency profiling and an ablation study.

Strengths

Technical novelty and innovation

- The paper thoughtfully integrates structural re-parameterization only where it is most beneficial (neck aggregation) rather than uniformly across all stages, which is a credible hardware-aware design choice.
- The use of a CoordConv stem to encode elevation priors for helmet-vs-floor disambiguation is a simple, targeted design that aligns with the task's spatial structure.
- Integrating BiFormer (bi-level routing attention) into the neck while preserving ONNX/TensorRT friendliness is a pragmatic contribution for applied deployment.

Experimental rigor and validation

- The authors present multi-platform latency profiling (PyTorch, ONNX Runtime, TensorRT; T4/RTX 3050/MX230/CPU), with Nsight-based reasoning about MAC/DRAM behavior and kernel launch counts.
- They include zero-shot transfer to an external dataset (HHW) and a 5-fold, scene-stratified CV on SHWD to address data leakage concerns.
- The ablation study examines component-wise contributions and reports multi-seed variance.

Clarity of presentation

- Structural re-parameterization mechanics are clearly explained with equations and a clean training-to-deploy narrative; CoordConv and BiFormer roles are well-motivated for this domain.
- The system-level pipeline and deployment stack are described in sufficient implementation detail to be informative for practitioners.

Significance of contributions

- The problem (real-time, reliable PPE monitoring) is practically important; the presented pipeline and cross-domain evaluation make this a

potentially useful engineering reference for the applied AI community.

⚠ Weaknesses

Technical limitations or concerns

- Methodological novelty is primarily in combining known components (RepConv, CoordConv, BiFormer, Focal-EIoU, TAL) rather than introducing a fundamentally new algorithmic idea.
- The approach remains NMS-based with DETR; it does not explore more



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Tech Overview

Experimental gaps or methodological issues

- In Table II (multi-seed, standardized SGD), the full system (A6) underperforms the baseline (A0) by ~ 1.2 mAP50 and ~ 1.24 mAP50-95 on average, contradicting the single-run AdamW “benchmark” improvement and raising concerns about robustness of the claimed accuracy gains.
- Cross-paper accuracy and FPS comparisons mix different training recipes, hardware, and precision modes; several baselines are taken from literature rather than re-trained under a matched protocol and runtime.
- The claimed per-class gains (notably for “hat”) are not consistently reconciled with joint mAP drops for “person,” and the paper does not report confidence intervals or statistical tests for these per-class outcomes.
- No Jetson or embedded NPU results are reported (despite “embedded devices” claims); MX230/CPU are helpful but not representative of typical modern edge AI accelerators used on sites.

Clarity or presentation issues

- The optimizer switch (AdamW for “benchmark” vs SGD for multi-seed) confounds interpretation; the paired t-test p-value is reported without stating the direction of effect, which, judging by the means, appears negative for A6 vs A0.

- The role/status of the “P2 Micro-Head” is unclear in the final configuration; it appears in ablations but not in the final fusion description.

Missing related work or comparisons

- Limited discussion/empirical comparison to modern NMS-free detectors and assignment/loss schemes such as YOLO26 (ProgLoss/STAL, DFL-free head) and RT-DETR variants that often excel at small objects on edge stacks.
- Related helmet-compliance literature on explicit person–helmet association and recent VLM-based zero-shot methods (e.g., OWLv2 pipelines) is briefly tangential in this paper; a more direct contrast on association or attribute-level evaluation would strengthen the positioning.

Detailed Comments

Technical soundness evaluation

- The structural re-parameterization derivation and folding of Conv+BN branches are standard and correctly specified. The reported reduction in memory traffic and kernel launches is plausible and consistent with prior RepVGG/Rep-based work.
- The CoordConv stem design is technically straightforward and well-justified for this task; the reported overhead (+0.06 ms) is negligible.
- The BiFormer integration and claims of ONNX/TensorRT compatibility via stock Ops (TopK, Gather, Gemm) are credible and practically relevant; however, details about memory fragmentation and batch-1 dynamic shape handling in TensorRT would be useful for replication.
- The Focal-EIoU + TAL combination is reasonable. The decoupling rationale for class imbalance is sound and aligns with established literature on assignment and loss balancing for small/rare classes (see Oksuz et al., 1909.00169).

Experimental evaluation assessment

- The cross-domain HHW zero-shot gains over the YOLO11s baseline are encouraging and align with the intuition that CoordConv (elevation priors) and token routing suppress cross-domain clutter.
- The 5-fold CV procedure is commendable for leakage control. However, the discrepancy between the single-run AdamW benchmark (showing small gains) and the standardized multi-seed SGD results ($A_6 < A_0$) should be resolved; as written, the evidence for a consistent accuracy improvement on SHWD is mixed.
- The ablations quantify latency impacts well (e.g., RepConv fusion -0.50 ms; BiFormer $+0.24$ ms), but the accuracy changes per component are very small and within tight SDs; more runs or CIs and per-class deltas would help interpret practical significance.
- Energy efficiency results are informative, but the absence of Jetson-class measurements is notable given the stated target of edge deployment.

Comparison with related work (using the summaries provided)

- Relative to BiFormer (2303.08810), the paper appropriately leverages routing attention for dense prediction efficiency and highlights deployment compatibility. A direct accuracy/latency contrast with a pure ViT/RT-DETR (e.g., 2604.23442) or TinyFormer (2605.25046) under matched conditions would contextualize performance on tiny objects.
- The choice to retain NMS and DFL contrasts with YOLO26's NMS-free and DFL-free design (2510.09653), which reportedly improves conversion/quantization stability and latency variance—highly relevant to edge stacks. A head-to-head with YOLO26 on SHWD/HHW would strengthen claims of deployment advantage.
- The structural re-parameterization positioning is coherent with RepCNN/RepNAS/RepGhost trends (2406.02652, 2109.03508, 2211.06088), but the paper would benefit from a short quantitative comparison or discussion of RepGhost/ODBB-like choices vs the current RepConv usage in the neck.
- From the hardhat detection angle, recent zero-shot VLM approaches (2410.12225) prioritize association and open-vocabulary flexibility; while the present work is supervised and single-task, a brief discussion of association (beyond the bipartite 0.25 IoU heuristic) and of deployment trade-offs vs VLM cascades would position the contribution more clearly.

Discussion of broader impact and significance

- Practically, the work addresses an important safety application and demonstrates a credible route to real-time multi-stream monitoring on commodity GPUs with a clear systems perspective (decode→preprocess→infer→NMS→render).
- The privacy-preserving on-premise deployment and simple post-hoc safeguards (helmet–person overlap filter) are appreciated. However, the association heuristic should be evaluated quantitatively, given its potential to reduce false positives at the cost of recall (e.g., helmets momentarily outside person boxes).
- The reported cross-domain robustness is valuable for real deployments, though broader validation (additional sites/datasets, seasonal shifts, camera angles) would better establish generalization.

🔍 Questions for Authors

1. The multi-seed standardized SGD results (A6) underperform the baseline (A0), yet the single-run AdamW “benchmark” shows a small gain. Can you reconcile these outcomes? Which optimizer/protocol should be considered primary evidence, and what is the direction of the paired t-test effect?
2. Please report per-class AP/AR and 95% CIs for SHWD across A0–A6, and clarify whether the person-class degradation is the main reason the joint mAP50-95 barely changes (or drops) despite helmet gains.
3. What is the final configuration regarding the P2 Micro-Head? It appears in ablations but not in the “full fusion” description. If excluded from the final model, why?
4. Can you provide matched, re-trained baselines (same data splits/augmentations/optimizer/hardware) for EC-YOLOv8/MAF-YOLO/TinyFormer or, alternatively, include YOLO26 and RT-DETR variants under an identical training and TensorRT FP16 export path?
5. Have you evaluated INT8 post-training quantization or QAT on TensorRT/CoreML/NCNN? If not, do your ONNX graphs (with BRA) convert cleanly to INT8 with calibration? Any observed accuracy/latency trade-offs?

6. Please provide a quantitative assessment of the person–helmet association heuristic (e.g., precision/recall changes vs no association and vs nested/cascaded baselines), ideally on a subset with reliable association labels.
7. How sensitive is BRA to S and top- k ($S=8$, $k=4$)? Can you share a small sweep (accuracy/latency) and whether routing instability appears on small batch or dynamic-resolution streams?
8. Could you share memory-bandwidth measurements beyond a single neck block (e.g., layer-wise Nsight CUPTI data), and clarify whether memory savings persist at higher resolutions (e.g., 1280^2)?

Overall Assessment

This is a well-executed applied paper with a clear deployment focus, sensible architectural choices, and extensive latency profiling. The integration of RepConv (neck-only), CoordConv stem, and BiFormer for small-object focus is coherent and well-motivated for helmet detection. The cross-domain HHW zero-shot improvements over a YOLO11s baseline are promising, and the systems-level details (ONNX/TensorRT compatibility, pipeline timing) make the work practically useful.

However, the central accuracy claims on SHWD are insufficiently substantiated: under standardized multi-seed SGD, the full system underperforms the baseline, while the single-run AdamW benchmark shows a slight improvement. This inconsistency, the reliance on mixed, non-matched baselines from the literature, and the absence of comparisons to modern NMS-free detectors (YOLO26/RT-DETR) weaken the evidence that the proposed combination offers a consistent accuracy–latency advance over strong contemporary alternatives. Methodological novelty is modest, as the paper primarily composes existing techniques.

For the AAIML Applied AI and Computer Vision Track, this submission is close: it offers solid engineering value and deployment insights but needs stronger, cleaner accuracy evidence under matched protocols and broader, fair comparisons. I recommend rejection in its current form, with a clear path to strengthening: unify training/evaluation protocols, resolve the A6 vs A0

inconsistency, add matched baselines (including YOLO26/RT-DETR), quantify the association heuristic, and consider embedded (Jetson) evaluation and INT8 results.

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